

genaut — frontier report

source CSV : genaut/logs/default.csv
corpus rows: 929

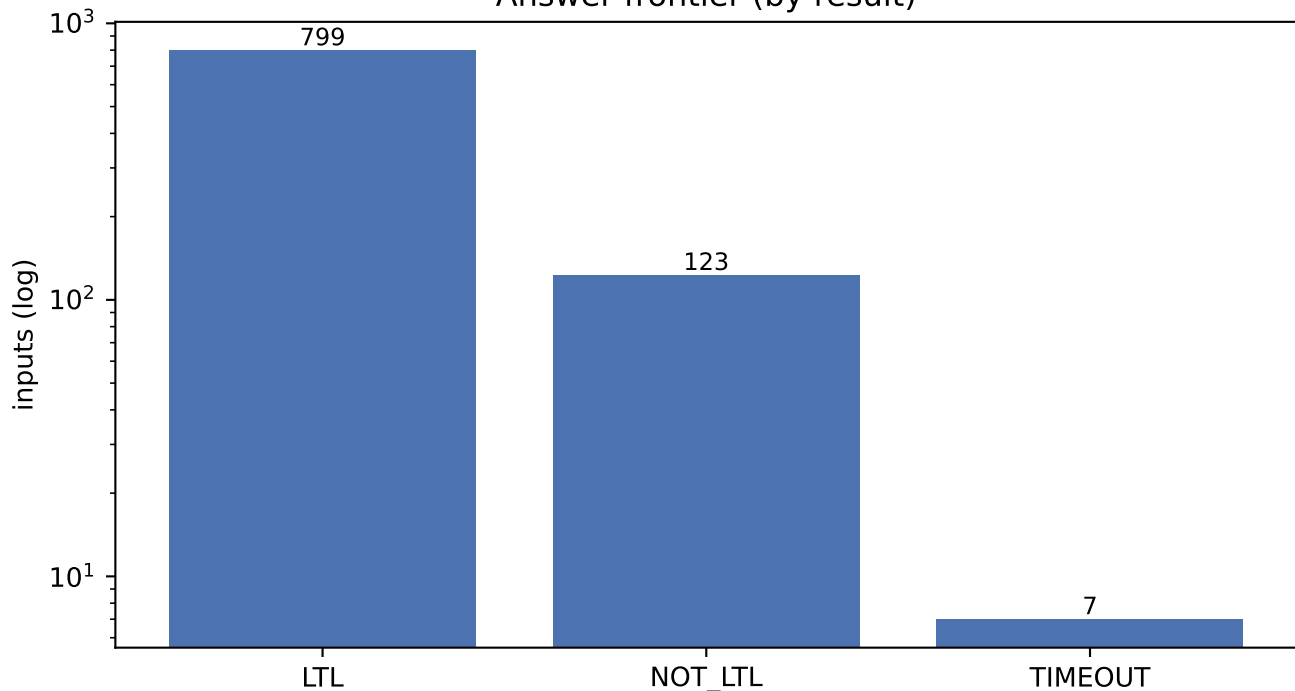
sections:

- Ventilation – LTL / not-LTL / ambiguous
- Formula collapse – idiom concentration
- Size frontier – flattened formula size
- Routes to `true` – which technique built the universal 1
- Build time & timeouts
- Verification – spot-oracle validation

Ventilation — LTL / not-LTL / ambiguous

```
corpus inputs: 929
  LTL-definable (formula built) :   799  ( 86.0%)
  not-LTL (decided)             :   123  ( 13.2%)
  ambiguous (timeout, unknown)  :     7  (  0.8%)
decided (LTL or not-LTL): 922  ( 99.2%)
```

Answer frontier (by result)



Formula collapse — idiom concentration

LTL answers: 799 (53 gated/unflattened strings)

distinct readable formulas: 87

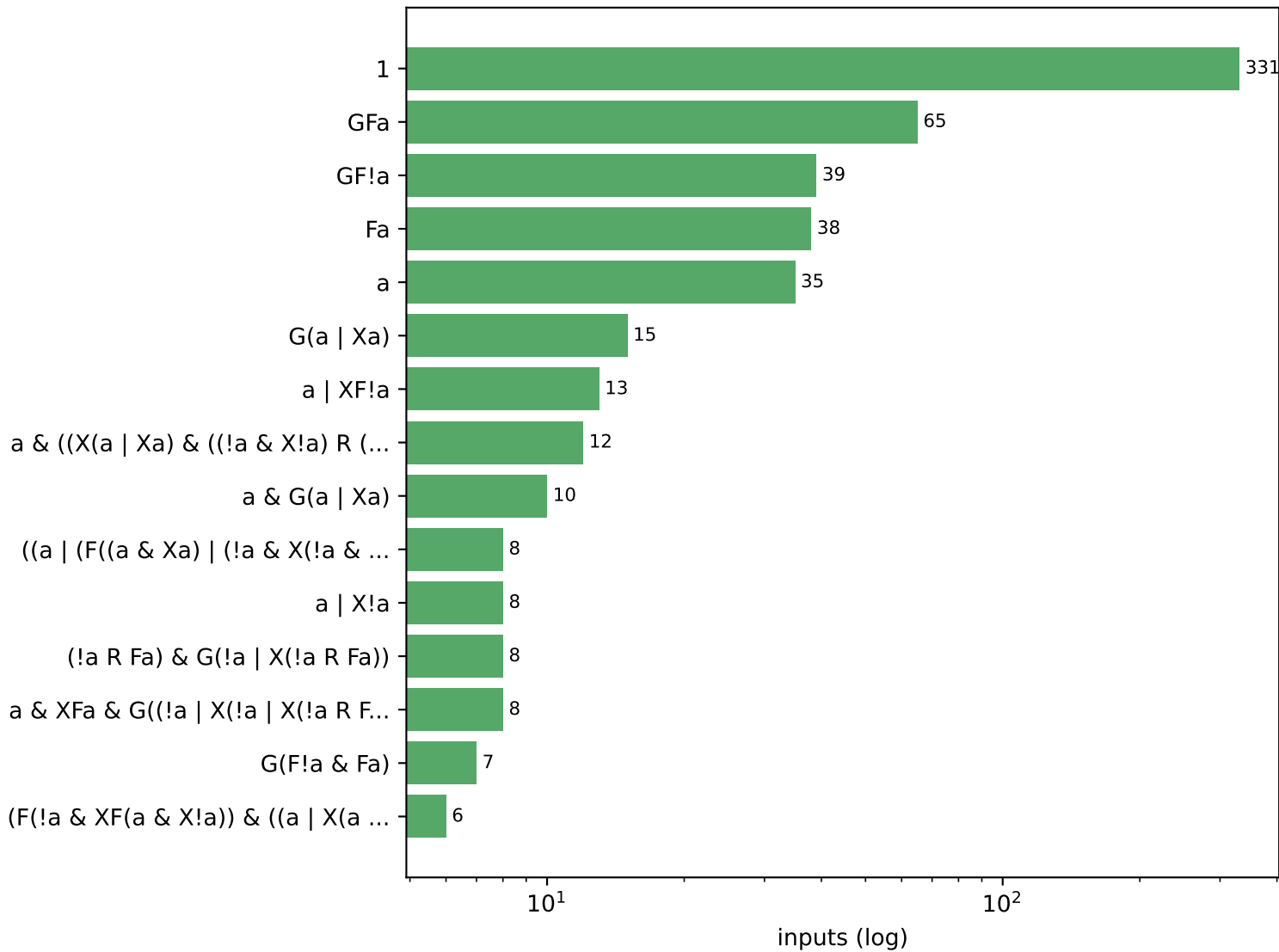
80% of readable answers covered by the top 14 idiom(s)

90% of readable answers covered by the top 32 idiom(s)

top 15 idioms:

331	(41.4%)	1
65	(8.1%)	GFa
39	(4.9%)	GF!a
38	(4.8%)	Fa
35	(4.4%)	a
15	(1.9%)	G(a Xa)
13	(1.6%)	a XF!a
12	(1.5%)	a & ((X(a Xa) & (!!a & X!a) R (!!a X(!a ...
10	(1.3%)	a & G(a Xa)
8	(1.0%)	((a (F((a & Xa) (!a & X(!a & XFa))) & (X!...
8	(1.0%)	a X!a
8	(1.0%)	(!a R Fa) & G(!a X(!a R Fa))
8	(1.0%)	a & XFa & G (!!a X(!a X(!a R Fa))) & (a ...
7	(0.9%)	G(F!a & Fa)
6	(0.8%)	(F(!a & XF(a & X!a)) & ((a X(a & XF!a)) U (...

Top 15 reconstructed idioms



Size frontier — flattened formula size

LTL answers with a size: 799

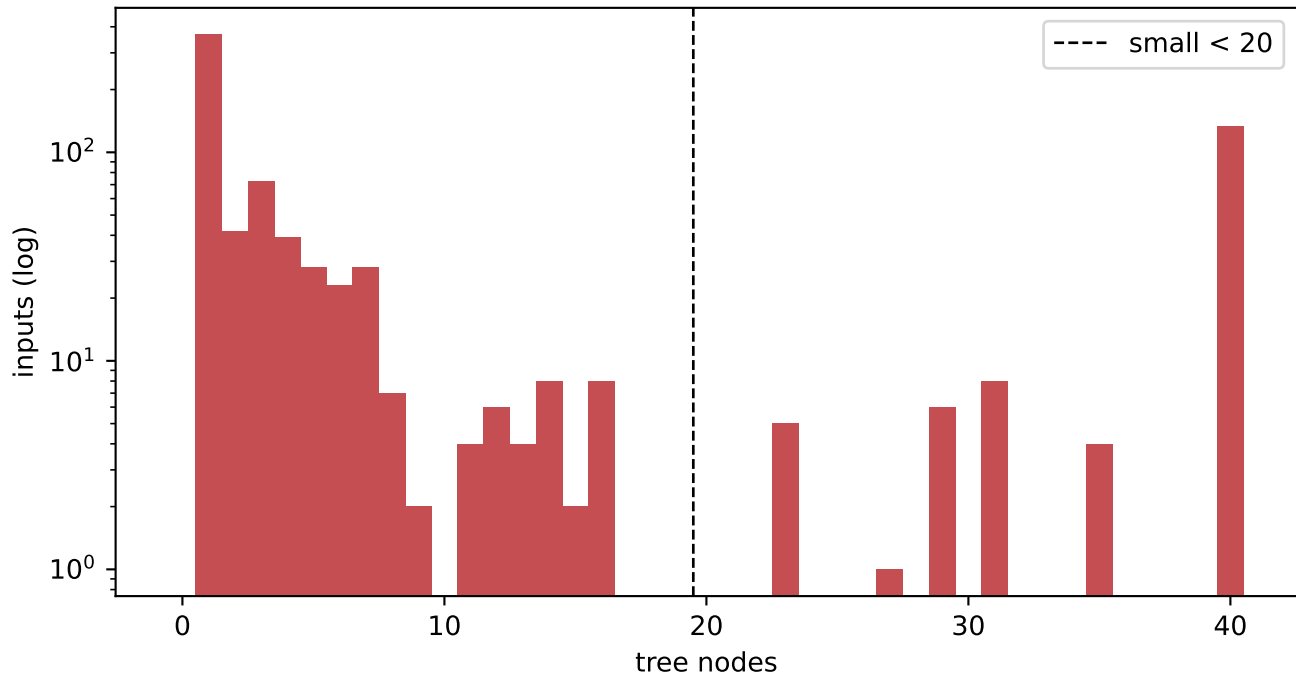
trivial (tree≤1, i.e. true/false): 367 (45.9%)

small (tree<20): 641 (80.2%)

larger (tree≥20): 158 (19.8%)

tree nodes: median 2 mean 132060661287776243759494201344.0 max 96278864367642945077320133640192

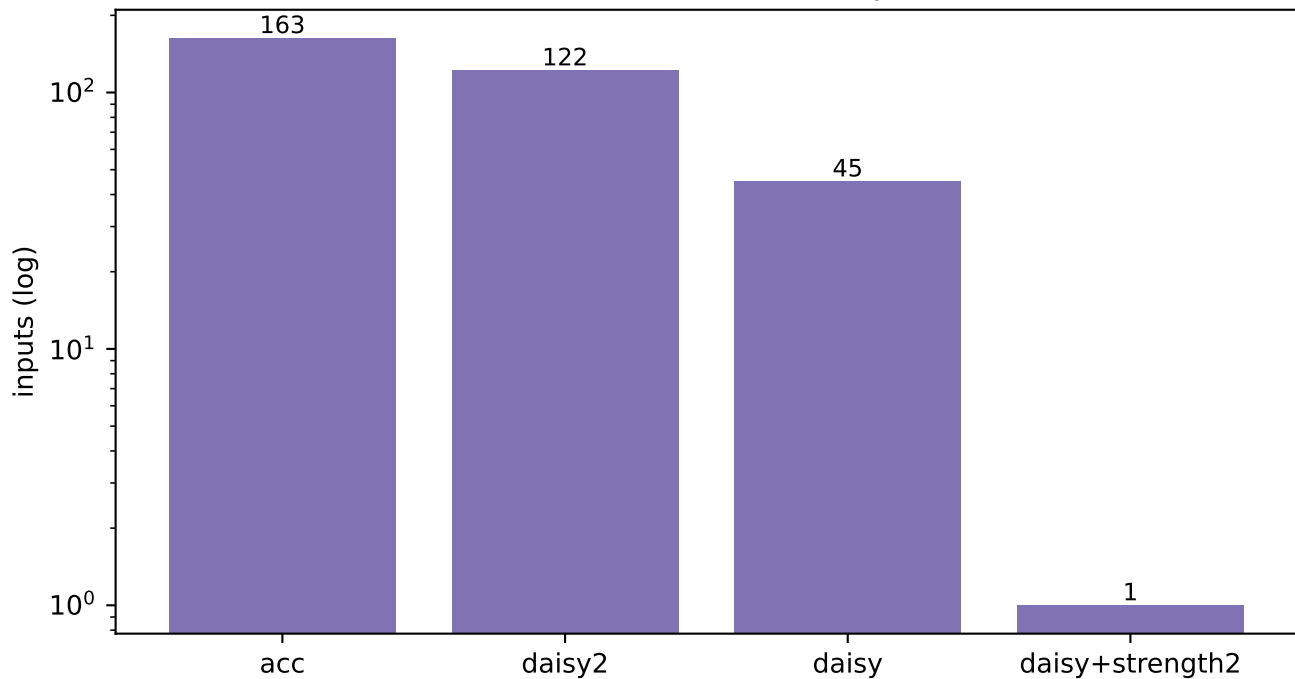
Flattened size (tree nodes); clipped at 40 (134 larger pooled at right)



Routes to `true` — which technique built the universal 1

```
universal answers (formula == '1'): 331
  163 ( 49.2%) acc
  122 ( 36.9%) daisy2
   45 ( 13.6%) daisy
    1 (  0.3%) daisy+strength2
build_s over true rows: total 191.5s  mean 0.579s  max 1.144s
```

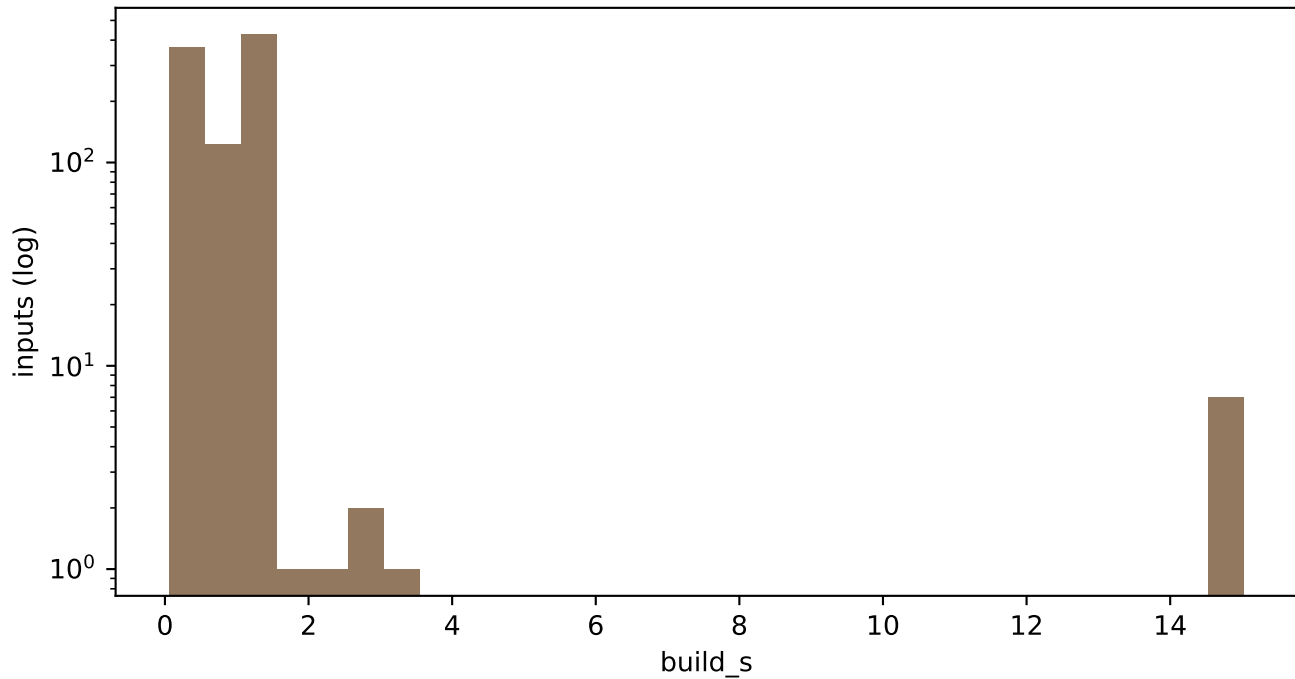

Routes to `true` (technique)



Build time & timeouts

```
total build: 693.9s    mean 0.747s    max 15.021s
  LTL          n= 799  mean 0.647s    max 3.101s
  NOT_LTL      n= 123  mean 0.583s    max 0.611s
timeouts: 7
  aut_09412.hoa
  aut_11460.hoa
  aut_18499.hoa
  aut_18563.hoa
  aut_33916.hoa
  aut_33986.hoa
  aut_34380.hoa
```

Build-time distribution



Verification — spot-oracle validation

```
validation over 799 LTL answers:  
  TRUE          745 ( 93.2%)  
  SIZE          53 (  6.6%)  
  TIMEOUT        1 (  0.1%)  
NOT-equivalent (FALSE): 0    CLEAN
```